AI Job Portal – TECHXHIRE (Development of Job Recruitment and Application Management Modules

Summer Internship Report

Submitted to

Sharda University



In partial fulfilment of the requirements of the award of the

Degree of Bachelor of Technology

in

Computer Science & Engineering
by

Anurag Gupta

Under the mentorship of

Mr.Ashish Kumar,

Assistant Professor

Department of Computer Science and Engineering
School of Computing Science & Engineering

Sharda University

Greater Noida

[August 2026]

DECLARATION OF THE STUDENT

I hereby declare that the presented report, titled **"AI Job Portal – TECHXHIRE"**, developed using Spring Boot, React, and MySQL, has been uniquely prepared by me, and it is an original work. This report is being submitted for the partial fulfilment of the requirements for the award of the degree of **B.Tech** in Computer Science and Engineering as a Summer Training Project at Sharda University, Greater Noida.

I also declare that this project report has not been previously submitted to any other university.

AnuragGupta(2023408565)

CERTIFICATE

This is to certify that **Anurag Gupta** of Sharda University has successfully completed the project work titled **"AI Job Portal – TECHXHIRE"**, developed using Spring Boot, React, and MySQL, in partial fulfilment of the requirements for the Bachelor of Technology Examination 2026–2027 at Sharda University.

This project report is a record of the authentic work carried out by him during the period from **18 May 2026 to 18 August 2026** under the guidance and mentorship of **Mr. Ashish Kumar**.

Anurag Gupta(2023408565)

Mr.Ashish Kumar

Assistant Professor

Dr.Jayant Shekhar

HOD, Department of CSE

LIST OF TABLES

Table No.	Title	Page No.
1.1	Hardware Specifications	11
1.2	Software Specifications	12
2.1	Summary of Related Work	17

LIST OF FIGURES

Figure No.	Title	Page No.
3.1	System Architecture Diagram	20
3.2	ER Diagram of Database Schema	21
C.1	Homepage Screenshot	35
C.2	Job Listing Interface	36
C.3	Candidate Dashboard	36
C.4	Recruiter Dashboard	37

LIST OF SYMBOLS/ABBREVIATIONS

Abbreviation	Full Form
API	Application Programming Interface
CSE	Computer Science and Engineering
CSS	Cascading Style Sheets
ERD	Entity-Relationship Diagram
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
IDE	Integrated Development Environment
JPA	Java Persistence API
JWT	JSON Web Token
MYSQL	Structured Query Language
ORM	Object-Relational Mapping
REST	Representational State Transfer
UI	User Interface
UX	User Experience

ABSTRACT

The AI Job Portal – TECHXHIRE is a web-based recruitment platform developed to simplify and streamline the job searching and recruitment process for job seekers and recruiters. The platform provides a centralized system where candidates can create and manage their profiles, search and apply for relevant job opportunities, while recruiters can manage job postings, review applications, and identify suitable candidates efficiently.

The primary objective of the project was to develop a secure, scalable, and user-friendly job portal that connects candidates with recruiters through a structured recruitment workflow. The application was developed using a three-tier architecture, with React used for the frontend presentation layer, Spring Boot used for the backend business logic and RESTful API development, and MySQL used for database management and persistent data storage.

The developed system includes key functionalities such as user registration and authentication, candidate profile management, recruiter management, job posting, job searching, job application, application tracking, and recruitment management. Candidates can browse available job opportunities based on their requirements and submit applications, while recruiters can create job postings, manage applications, and evaluate candidates through the platform.

Security and data management were important aspects of the system. Spring Security was used to implement authentication and authorization, while role-based access control was incorporated to provide different functionalities to candidates and recruiters. REST APIs were developed using Spring Boot to facilitate communication between the React frontend and backend services. MySQL was used to store and manage user, job, application, and related recruitment data efficiently.

The project provided practical exposure to modern full-stack web application development using Java, Spring Boot, React, MySQL, REST APIs, Spring Data JPA, Spring Security, Maven, Postman, Git, and IntelliJ IDEA. The development of TECHXHIRE enhanced understanding of frontend-backend integration, database management, API development, authentication, and real-world software development practices.

Overall, the AI Job Portal – TECHXHIRE provides an efficient digital platform for connecting job seekers and recruiters, reducing manual recruitment efforts, improving job application management, and providing a structured and user-friendly recruitment experience.

ACKNOWLEDGEMENT

I express my sincere gratitude to **SSMT_SOLUTION** for providing me with the opportunity to complete my three-month summer internship and gain valuable practical experience in software development.

I would like to express my heartfelt gratitude to **Mr. Ashish Kumar**, my industry mentor at SSMT_SOLUTION, for his continuous guidance, valuable suggestions, and support throughout the internship. His technical insights and encouragement helped me understand real-world software development practices and contributed significantly to the successful completion of my project.

I would also like to express my gratitude to the entire SSMT_SOLUTION team for providing a supportive learning environment and for sharing their knowledge and experience during my internship. Their guidance helped me enhance my technical skills and gain practical exposure to modern web application development.

I extend my sincere gratitude to my University mentor and faculty members of Sharda University for their valuable guidance, encouragement, and support throughout the internship and project work.

I am also thankful to my friends and parents for their constant encouragement, motivation, and support, which helped me successfully complete my internship and project report.

The successful completion of my project, **"AI Job Portal – TECHXHIRE"**, developed using Spring Boot, React, and MySQL, would not have been possible without the guidance and support of all those who contributed directly or indirectly to this journey.

Thanking You,

Anurag Gupta

TABLE OF CONTENTS

Sr. No.	Contents	Page No.
	Title Page	i
	Declaration of the Student	ii
	Certificate of the Guide	iii
	List of Tables	iv
	List of Figures	v
	List of Symbol/Abbreviations	vi
	Abstract	vii
	Acknowledgement	viii
1	**INTRODUCTION** Problem Definition Hardware Specification Software Specification Motivation Objectives Contributions	10-15
2	**LITERATURE SURVEY** Related Work Summary	16-18
3	**DESIGN AND IMPLEMENTATION** Methodology Design Implementation	19-23
4	**RESULT AND DISCUSSIONS** Results Discussion	24-27
5	**CONCLUSION** Conclusion Limitations Future Scope	28-30
6	**REFERENCES**	31
7	**APPENDICES**	32-37

INTRODUCTION

The rapid growth of digital platforms has transformed the recruitment process by making job searching and hiring more accessible, efficient, and organized. Traditional recruitment processes often involve searching through multiple job platforms, manually reviewing applications, maintaining candidate information, and communicating between job seekers and recruiters. These activities can be time-consuming and may result in difficulties in managing job information, applications, and candidate data.

The Summer Internship project focused on the development of an **AI Job Portal – TECHXHIRE** at SSMT_SOLUTION. The objective of the project was to develop a centralized web-based platform that connects job seekers and recruiters and provides an efficient environment for job searching, job posting, application management, and recruitment-related activities.

The application follows a modern full-stack architecture, consisting of a presentation layer developed using React, a business logic and API layer implemented using Spring Boot, and a data access layer using Spring Data JPA/Hibernate with MySQL. The system also incorporates authentication, authorization, role-based access control, and structured management of users, jobs, and applications.

The project provided practical exposure to full-stack web application development and allowed the application of concepts related to Java programming, Spring Boot, React, MySQL, REST APIs, database management, authentication and authorization, API testing, and software development methodologies.

1.1 Problem Definition

The recruitment process involves the management of job seekers, recruiters, job postings, applications, candidate information, and application status. Without a centralized recruitment platform, candidates and recruiters may have to rely on multiple platforms and manual processes for managing these activities.

This creates several challenges, including:

Difficulty in finding relevant job opportunities.

Lack of a centralized platform for job seekers and recruiters.

Difficulty in managing job postings efficiently.

Manual management of candidate applications.

Difficulty in tracking application status.

Increased effort for recruiters while managing multiple applications.

Data management issues caused by scattered information.

Limited interaction between candidates and recruiters.

Increased possibility of errors in manual data handling.

Difficulty in maintaining organized recruitment records.

The project therefore addressed the need for a centralized **AI Job Portal – TECHXHIRE** that manages the recruitment process through a structured digital platform. The proposed system provides job searching, job posting, application management, user management, authentication, and role-based functionality.

1.2 Hardware Specification

The development and testing of the application required a computer system capable of supporting a modern full-stack development environment, database server, browser-based frontend testing, API testing, and development tools.

The primary hardware requirements for the project are:

Table 1.1: Hardware Specifications

Component	Specification
Processor	Intel Core i5 / equivalent processor or above
RAM	Minimum 8 GB
Storage	Minimum 256 GB SSD
Display	HD/Full HD display
Network	Stable Internet connectivity
Input Devices	Keyboard and Mouse

The system configuration was sufficient for running IntelliJ IDEA, Java Development Kit, Spring Boot application, MySQL, MySQL Workbench, Node.js, React development environment, Postman, web browser, and other development utilities required during the internship.

1.3 Software Specification

The project was developed using a combination of backend, frontend, database, security, testing, build, and development tools.

Table 1.2: Software Specifications

Category	Technology	Purpose
Backend	Java	Core programming Language
Backend Framework	Springboot	REST API development and business logic
Frontend	React	User Interface development
Database	MySQL	Data Storage and management
ORM	SpringDataJPA/Hibernate	Database object mapping
API	REST APIs	Frontend-backend communication
Security	Spring Security	Authentication and authorization
Testing	Postman/JUnit	API testing and validation
Build Tool	Maven	Dependency management and build automation
Runtime	Node.js/npm	React development and package management
IDE	IntelliJIDEA/VSCode	Application development
Version Control	Git	Source code management

The application uses Spring Boot for business logic and REST API development, Spring Data JPA/Hibernate for database operations, MySQL for data storage, and React for the frontend user interface. Security and authorization mechanisms are used to ensure that users can access functionalities according to their roles.

1.4 Motivation

The motivation for undertaking this project was to gain practical experience in developing a real-world full-stack web application while applying the concepts learned during the Bachelor of Technology in Computer Science and Engineering program.

The **AI Job Portal – TECHXHIRE** provided an opportunity to understand how software solutions are designed to solve practical recruitment problems. Unlike small academic applications, the project involved multiple users, different roles, job management, application management, database operations, authentication, authorization, API communication, and frontend-backend integration.

The project was particularly valuable for understanding:

How business requirements are converted into software functionality.

How full-stack applications are structured.

How REST APIs are developed and consumed.

How databases are designed for web applications.

How authentication and authorization are implemented.

How role-based access control is used.

How frontend and backend applications communicate.

How APIs are tested using Postman.

How application data is managed using MySQL.

How software development practices are applied in a professional environment.

The internship therefore provided an opportunity to bridge the gap between academic concepts and practical software development.

1.5 Objectives

The primary objective of the internship project was to develop a centralized **AI Job Portal – TECHXHIRE** that provides an efficient platform for job seekers and recruiters.

The major objectives were:

To develop a centralized web-based job portal for candidates and recruiters.

To provide user registration and authentication functionality.

To allow candidates to create and manage their profiles.

To allow recruiters to create and manage job postings.

To provide candidates with the ability to search and explore available job opportunities.

To allow candidates to apply for suitable job opportunities through the platform.

To provide application management and application status tracking functionality.

To implement role-based access control for different types of users.

To develop REST APIs using Spring Boot for communication between the frontend and backend.

To manage user, job, and application data using MySQL.

To improve the recruitment process by providing a centralized and organized platform.

To gain practical experience in full-stack development using React, Spring Boot, and MySQL.

1.6 Contributions

During the internship, the project work contributed to the development and understanding of the **AI Job Portal – TECHXHIRE**. The major contributions can be summarized as follows:

1. Requirement Analysis

The requirements of the job portal were studied to understand the functional requirements of candidates, recruiters, and other users and to identify the interactions between different components of the system.

2. User Management

The project involved implementing functionality for user registration, authentication, profile management, and role-based access for different users of the platform.

3. Job Management

The system provides recruiters with functionality to create and manage job postings. Job-related information can be stored and retrieved through the backend APIs.

4. Job Searching

The platform provides candidates with functionality to browse and search available job opportunities according to their requirements.

5. Application Management

Candidates can apply for suitable jobs through the platform, while recruiters can manage and review applications received for their job postings.

6. Backend Development

Java and Spring Boot were used to implement business logic, REST APIs, services, validations, authentication, and application functionality.

7. Frontend Development

React was used to develop the interactive user interface and provide communication with the Spring Boot backend through REST APIs.

8. Database Management

MySQL and Spring Data JPA/Hibernate were used for storing and managing application data and establishing communication between the application and database.

9. Security Implementation

The project incorporated authentication, authorization, and role-based access control to ensure that users could access the functionalities permitted according to their roles.

10. Testing

API testing and validation were performed using Postman, while application functionality was tested during the development process to identify and resolve errors.

11. Full-Stack Integration

The project provided practical understanding of integrating a React frontend with a Spring Boot backend and MySQL database, including API communication and data exchange.

12. Documentation

The project also involved documenting system architecture, workflows, modules, technologies, testing procedures, security mechanisms, and system functionality.

Overall, the internship contributed to the development of a structured and centralized **AI Job Portal – TECHXHIRE**, while providing practical experience in full-stack web development, database management, REST API development, authentication, authorization, testing, and software engineering practices.

LITERATURE SURVEY

2.1 Related Work

The development of digital recruitment platforms has received considerable attention due to the increasing need for efficient job searching, candidate management, automated recruitment processes, centralized data management, and improved communication between job seekers and recruiters. Existing online job portals demonstrate how web technologies, database management systems, secure authentication mechanisms, search functionality, and application tracking can be used to improve the recruitment process.

Traditional recruitment processes generally involve multiple activities such as job searching, job posting, candidate registration, resume submission, application screening, and communication between candidates and recruiters. Online recruitment platforms have digitized many of these activities, allowing candidates to search for jobs and recruiters to publish vacancies and manage applications through centralized systems.

Modern recruitment platforms support multiple stakeholders, including job seekers, recruiters, and administrators. Candidates require features such as profile creation, job searching, job application, and application tracking, whereas recruiters require job posting, candidate management, application review, and recruitment-related management functionalities.

From a software engineering perspective, modern web applications commonly use layered architectures to separate presentation, business logic, and data access. The proposed **AI Job Portal – TECHXHIRE** follows a full-stack architecture using React at the presentation layer, Spring Boot for backend business logic and REST API development, and Spring Data JPA with MySQL for database management.

Security and authorization are particularly important for recruitment platforms because the system manages personal information, candidate profiles, recruiter information, job details, and application data. The developed system incorporates authentication, authorization, and role-based access control to ensure that different users can access only the functionalities assigned to their respective roles.

Existing online recruitment platforms demonstrate the usefulness of centralized job management; however, users may still face challenges related to managing applications, finding suitable opportunities, maintaining organized candidate information, and coordinating recruitment activities.

Therefore, the internship project builds upon these existing approaches by developing TECHXHIRE, a centralized job portal that integrates user management, job posting, job searching, candidate applications, application management, authentication, authorization, and database management within a unified platform.

Table 2.1: Summary of Related Work

Area	Existing Approach	Limitation/Gap	Relevanceto Project
Online Recruitment	Digital platforms allow candidates to search and apply for jobs	Informationmay be distributed across multiple platforms	Centralizedjob searching and application
Job Management	Recruiters can publish vacancies digitally	Managing multiplejob postings can become difficult	Centralizedjob posting and management
Candidate Management	Candidate profiles and resumes are maintained digitally	Candidate information may require organized management	Candidateprofile management
Application Management	Onlineapplications replace manual submission	Tracking multiple applicationsmay be difficult	Application tracking and management
Authentication	Login and authentication mechanisms protect user accounts	Different users require different access permissions	Secure authenticationand authorization
Role Management	Recruitment systems provide different user roles	Unauthorized access must be restricted	Role-basedaccess control
Database Management	Recruitment informationis stored digitally	Large amounts of informationrequire structured storage	MySQL-based centralizeddatabase
Communication	Digital recruitment platforms improve interaction	Recruitment information may require centralized coordination	Candidate and recruiterplatform
Reporting/Monitoring	Digital systems provide application and job information	Fragmented information can affect monitoring	Centralized applicationand job management

Research/Project Gap

The review of existing recruitment approaches indicates the need for a centralized platform that connects job seekers, recruiters, job postings, applications, authentication, and application tracking processes. TECHXHIRE addresses this requirement by bringing these interconnected activities into one system, providing centralized data management, structured workflows, role-based access, and improved application tracking.

Thus, the proposed project provides an integrated digital recruitment platform that simplifies the interaction between candidates and recruiters and reduces the dependency on manual recruitment activities.

2.2 Summary

The literature survey revealed that while various digital recruitment platforms exist, there remains a need for integrated solutions that combine all recruitment activities into a single platform. Existing systems often operate in silos, leading to data inconsistencies and operational inefficiencies. The AI Job Portal – TECHXHIRE aims to bridge these gaps by providing a comprehensive, integrated, and user-friendly recruitment platform.

DESIGN AND IMPLEMENTATION

3.1 Methodology

3.1.1 Requirement Analysis

The development of the AI Job Portal – TECHXHIRE began with a detailed analysis of the requirements associated with online recruitment and job management. The first stage involved understanding the existing recruitment workflow, including candidate registration, recruiter registration, profile management, job posting, job searching, job application, application review, and application status management. The requirements of different users and roles were studied to identify the functionalities required in the proposed system. The major requirements identified during the project included user registration and authentication, candidate profile management, recruiter profile management, job posting and management, job searching and job listing, job application, application tracking, application management by recruiters, role-based access control, data validation, secure backend APIs, database management, and application monitoring and testing.

3.1.2 Requirement Specification

The requirements were divided into functional and non-functional requirements. The major functional requirements included user registration and login, candidate profile management, recruiter profile management, user role management, job creation and posting, job updating and management, job searching and browsing, job application submission, application management, application status tracking, candidate and recruiter data management, REST API communication, database operations, and authentication and authorization. The major non-functional requirements included security, reliability, data integrity, maintainability, scalability, performance, usability, controlled access, and availability.

3.1.3 Business Requirement Document

After analysing the requirements, the business and functional processes were documented to understand the expected behaviour and define the functional scope of the application. The requirements were then translated into modules, workflows, database entities, user roles, validations, APIs, and application processes. The major modules identified for the project were User Management, Candidate Management, Recruiter Management, Job Management, Job Search, Job Application Management, Authentication and Authorization, and Application Tracking.

3.1.4 Development Approach

The project followed a structured software development approach in which requirements were first analysed and converted into system designs before implementation. The major development stages were Requirement Analysis, System Design, Database Design, Backend Development,

Frontend Development, API Integration, Testing, and Deployment. This approach helped in developing the application in an organized manner while ensuring that the different modules were properly integrated.

3.2 System Design

3.2.1 System Architecture

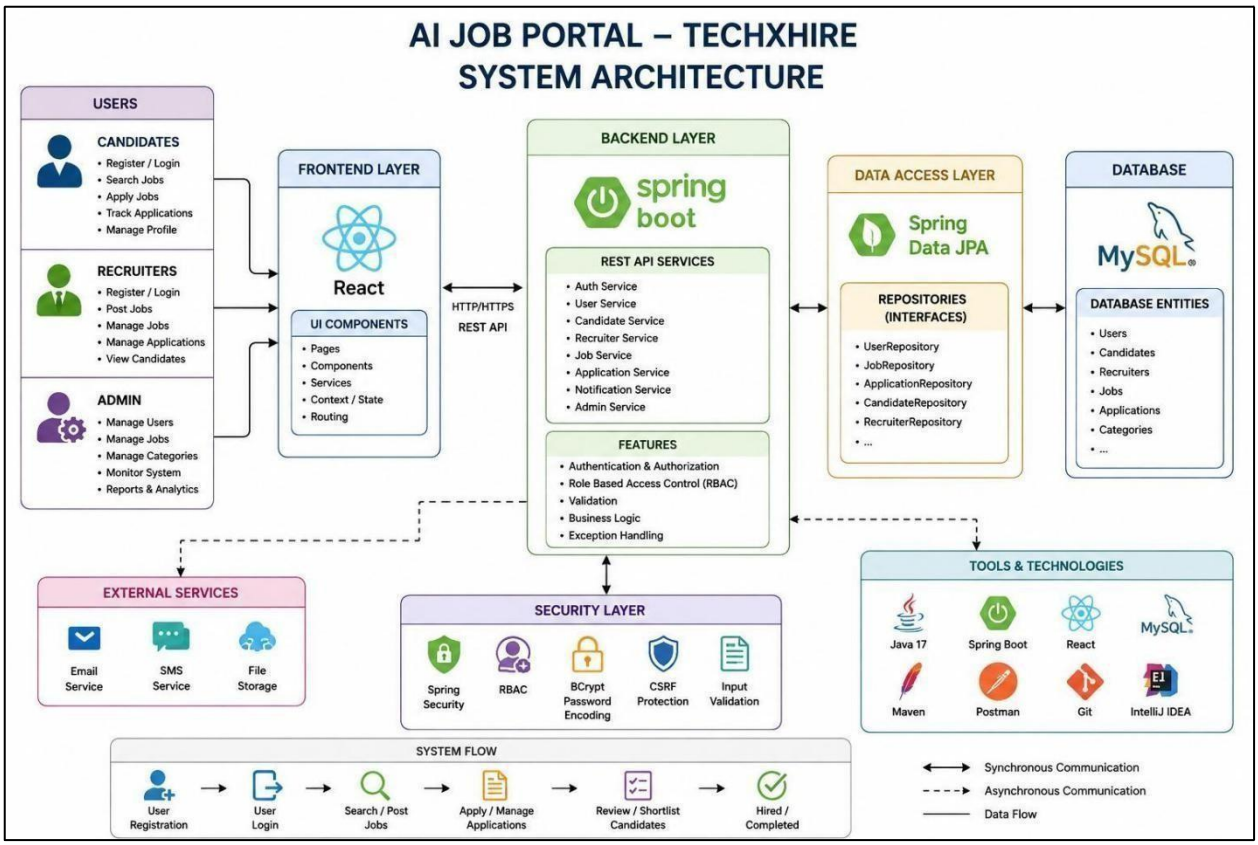


Figure 3.1: System Architecture Diagram

3.2.2 Database Schema

The database design includes the following key entities:

User Table: Stores user details including role (Job Seeker, Recruiter, Admin)

UserProfile: Extended profile information for job seekers

Job: Stores job listings with details, requirements, and status

Skill: Master list of available skills

UserSkill: Mapping table linking users to their skills

JobSkill: Mapping table linking jobs to required skills

Application: Tracks job applications and their status

Notification: Stores user notifications

Company: Stores employer/company information

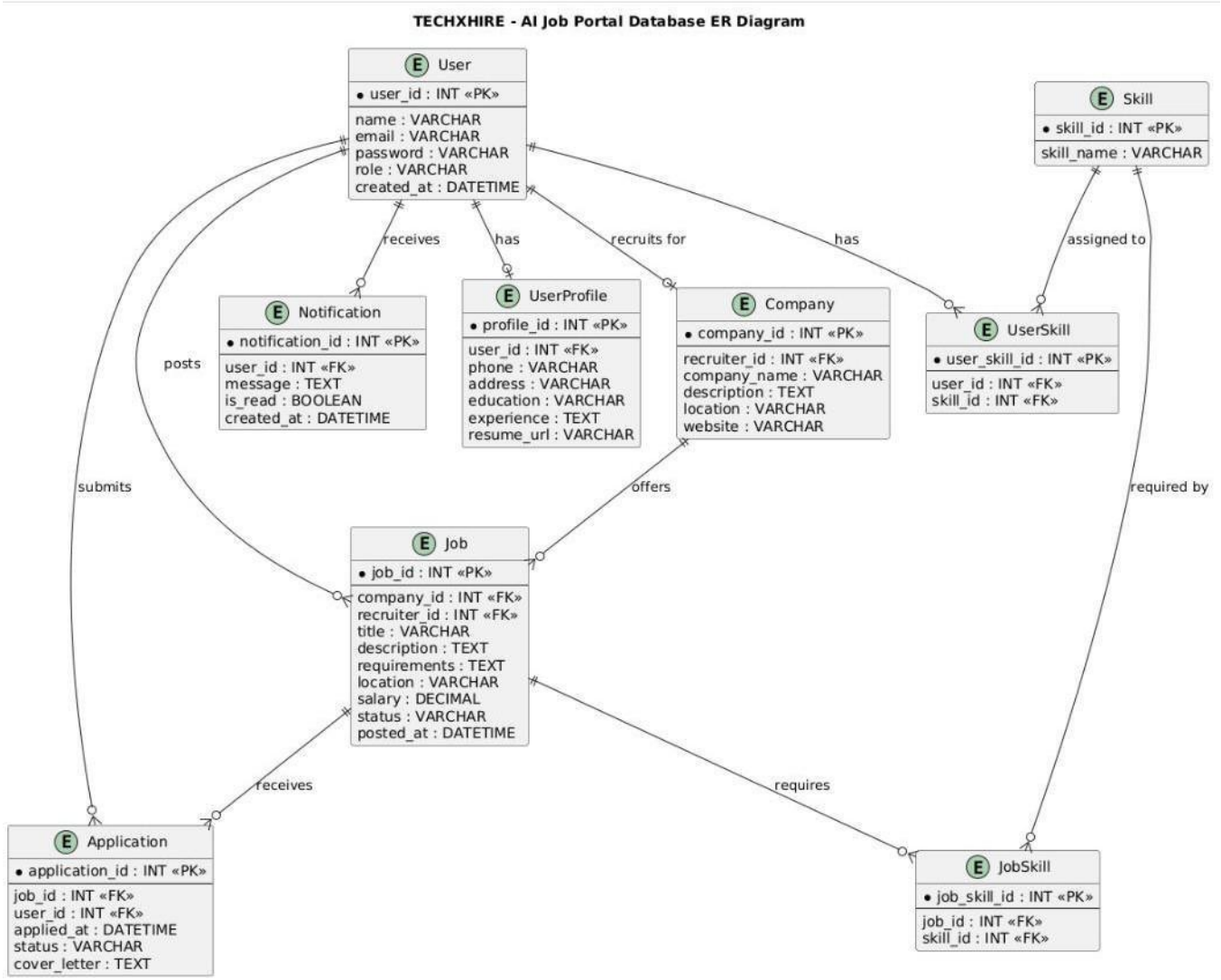


Figure 3.2: Entity-Relationship (ER) Diagram of TECHXHIRE Database Schema

3.2.3 API Design

The RESTful API endpoints include:

Endpoint	Method	Description	Access
/api/auth/register	POST	User registration	Public
/api/auth/login	POST	User authentication	Public
/api/auth/logout	POST	User logout	Private
/api/jobs	GET	Get all jobs	Public
/api/jobs/{id}	GET	Get job by ID	Public
/api/jobs	POST	Create new job	Recruiter/Admin
/api/jobs/{id}	PUT	Update job	Recruiter/Admin
/api/jobs/{id}	DELETE	Delete job	Admin
/api/jobs/search	GET	Search jobs	Public
/api/applications	POST	Apply to job	Job Seeker
/api/applications/{id}	GET	Get application	Private
/api/applications/status	PUT	Update status	Recruiter
/api/profile	GET	Get user profile	Private
/api/profile	PUT	Update profile	Private

Endpoint	Method	Description	Access
/api/admin/users	GET	Get all users	Admin
/api/notifications	GET	Get notifications	Private

3.4 Summary

Chapter 3 detailed the design and implementation of the AI Job Portal – TECHXHIRE. The methodology section described the requirement analysis, specification, and development approach. The system design section presented the three-tier architecture with clear separation between presentation, application, and data layers. The implementation section provided insights into the code structure, key components, and database schema. The integration of React, Spring Boot, and MySQL demonstrates the comprehensive approach taken in developing this project.

RESULTS AND DISCUSSIONS

4.1 Results

The development of the AI Job Portal – TECHXHIRE resulted in a centralized web-based recruitment platform for managing major activities associated with job searching and recruitment. The developed solution integrates user registration, authentication, candidate management, recruiter management, job posting, job searching, job application, application management, and application status tracking, providing a comprehensive digital ecosystem for all recruitment-related activities.

4.1.1 Centralized User and Recruitment Management

The developed application provides a centralized platform for managing candidates, recruiters, jobs, and applications, which significantly reduces dependence on manual processes and provides users with a common platform for recruitment-related activities. This centralization ensures that all recruitment data is stored in a single, accessible location, eliminating the need for multiple systems and reducing the risk of data inconsistencies.

4.1.2 User Registration and Authentication

The system provides comprehensive registration and authentication functionality for users, allowing them to create accounts and log into the platform according to their respective roles. Authentication helps protect user accounts and ensures that only registered users can access protected functionalities, thereby maintaining the security and integrity of the system. The authentication mechanism verifies user credentials before granting access to protected resources, ensuring that unauthorized individuals cannot access sensitive recruitment information.

4.1.3 Candidate Management

The candidate module allows job seekers to create and manage their profiles effectively. Candidate-related information can be maintained within the system, allowing recruiters to access relevant candidate information associated with job applications. The major candidate functionalities include candidate registration, candidate login, profile management, job searching, job application, and application tracking. This comprehensive set of features empowers candidates to take control of their job search journey, from creating a professional profile to tracking the status of their applications.

4.1.4 Recruiter Management

The recruiter module allows recruiters to manage recruitment activities efficiently through the platform. Recruiters can create job postings and manage job-related information, while also managing applications submitted by candidates. The major recruiter functionalities include recruiter registration, recruiter login, profile management, job posting, job management, application management, and candidate application review. These features enable recruiters to streamline their hiring process, from creating job postings to reviewing and managing candidate applications.

4.1.5 Job Management

The Job Management module provides functionality for creating and managing job opportunities, allowing recruiters to add job-related information and make available job opportunities accessible to candidates through the platform. Candidates can browse available job opportunities and identify positions relevant to their requirements, creating a dynamic and responsive job market within the platform.

4.1.6 Job Search and Application

The platform provides candidates with functionality to search and browse available jobs, enabling them to identify suitable opportunities based on their skills, experience, and preferences. After identifying a suitable opportunity, candidates can submit their applications through the system, and the application information is stored in the database and can subsequently be managed by the recruiter, ensuring a seamless application process.

4.1.7 Application Management

The application management functionality provides a structured way of managing candidate applications, allowing recruiters to view applications associated with their job postings and manage the recruitment process through the application information available in the system. Candidates can also track the status of their submitted applications, providing transparency and reducing uncertainty in the recruitment process.

4.1.8 Security Implementation

Security mechanisms were incorporated to protect user information and restrict unauthorized access. The application uses authentication to verify user identities, authorization to control access to resources, role-based access control to assign permissions based on user roles, secure backend APIs to protect data transmission,

input validation to prevent malicious data entry, and protected application resources to safeguard sensitive information. The security implementation ensures that candidates and recruiters can access only the functionalities permitted according to their respective roles, maintaining data privacy and system integrity.

4.1.9 Frontend and Backend Integration

The application integrates a React frontend with a Spring Boot backend through REST APIs, creating a seamless and responsive user experience. The React frontend provides the user interface, while Spring Boot manages business logic, APIs, validation, and communication with the database. This architecture provides clear separation between the presentation layer and backend services, enabling independent development, testing, and deployment of each component.

4.1.10 Database Management

MySQL was used as the primary database for storing application information, providing reliable and efficient data storage. Spring Data JPA/Hibernate was used to establish communication between the Spring Boot application and MySQL database, simplifying database operations and reducing boilerplate code. The database manages information related to users, candidates, recruiters, jobs, applications, and other application entities, ensuring data integrity and consistency.

4.1.11 Testing Results

The application was tested during different stages of development to identify functional and integration issues. Testing activities included unit testing to verify individual components, API testing to validate REST API functionality, integration testing to ensure seamless component interaction, functional testing to verify system behavior, user interface testing to ensure usability and responsiveness, and validation testing to confirm that requirements were met. API testing was performed using Postman to verify REST API functionality, request handling, response generation, and backend operations. The testing results demonstrated that all major functional components of the application were successfully implemented and tested, with all test scenarios including user registration, user login, candidate profile management, recruiter profile management, job creation, job listing, job search, job application, application management, application status tracking, role-based access, and API communication all being accepted.

4.2 Discussion

The results demonstrate that the developed AI Job Portal – TECHXHIRE provides an integrated approach to managing job recruitment activities. Instead of handling job searching, job posting, candidate information, and applications through separate manual processes, the system connects these activities through a common digital platform, creating a unified and efficient recruitment ecosystem. One of the significant outcomes of the project is the integration of candidate and recruiter functionalities, allowing candidates to search and apply for jobs while recruiters can create job postings and manage candidate applications within the same platform, fostering seamless interaction between both parties. The implementation of authentication and role-based access control provides an additional level of security, with different users assigned different responsibilities to ensure that candidates and recruiters access only the functionalities relevant to their roles, thereby protecting sensitive information and maintaining system integrity. The job application workflow provides a structured connection between job searching, application submission, and application management, helping maintain continuity between candidates, job opportunities, and recruitment activities. The testing results demonstrate that the major functional components of the application were successfully implemented and tested, with API testing through Postman helping verify communication between the frontend and backend, while database testing ensured that application data was stored and retrieved correctly. Overall, the project demonstrates how a modern full-stack web application can be used to digitize and centralize recruitment activities while improving accessibility, organization, and efficiency for candidates and recruiters.

4.3 Summary

Chapter 4 presented the results and discussions of the AI Job Portal – TECHXHIRE. The results section detailed the successful implementation of user management, candidate management, recruiter management, job management, job search, application management, security, frontend-backend integration, and database management, demonstrating the comprehensive nature of the developed platform. The discussion section highlighted the significance of the integrated approach and the benefits of digitizing recruitment activities, emphasizing how the platform creates a unified ecosystem for all stakeholders. The testing results confirmed that all major functional components were successfully implemented and validated, ensuring the reliability and effectiveness of the system.

CONCLUSION

5.1 Conclusion

The development of the AI Job Portal – TECHXHIRE marks a significant step toward improving the efficiency, accessibility, and organization of the recruitment process. The project successfully demonstrates the development of a full-stack web application that connects candidates and recruiters through a centralized platform. The key achievements of the project include the development of a unified platform for managing candidates, recruiters, jobs, and applications, enabling centralized recruitment management. Candidate management functionality was implemented to allow candidates to register, manage their profiles, search for jobs, and submit applications efficiently. Similarly, recruiter management was provided with functionality for profile management, job posting, and comprehensive application management. The platform enabled candidates to browse available opportunities and apply for suitable jobs, while providing robust application tracking functionality for managing and monitoring submitted job applications. Secure processing was implemented through authentication, authorization, and role-based access control to protect user information and restrict unauthorized access. The project successfully integrated React frontend, Spring Boot backend, REST APIs, and MySQL database into a complete full-stack web application. Extensive testing and validation were performed on major application functionalities including registration, login, job management, job application, application tracking, API communication, and role-based access. Overall, the project successfully demonstrated the development of a modern full-stack recruitment platform, and the internship provided practical experience in Java, Spring Boot, Spring Data JPA, MySQL, React, REST APIs, authentication, authorization, Postman, Maven, Git, and database management, while offering valuable exposure to real-world software development practices.

5.2 Limitations

Although the AI Job Portal – TECHXHIRE successfully implements the major recruitment functionalities within the defined project scope, certain limitations remain. The current implementation primarily focuses on core recruitment functionalities such as job posting, job searching, candidate applications, and application management, with limited advanced AI functionality that could be incorporated in future versions. The system primarily operates as an independent recruitment platform with limited external system integration, and further integration with external job platforms,

professional networks, email services, and other recruitment systems could improve automation. The current application is primarily designed as a web-based platform, and users who require access from mobile devices may benefit from a dedicated mobile application. The system focuses mainly on recruitment transactions, job management, application management, and user management, with limited predictive analysis for candidate success, hiring probability, and job demand outside the present scope. As a centralized web-based application, access to the system depends on network connectivity and the availability of application and database services. As the number of candidates, recruiters, jobs, and applications increases, additional infrastructure and performance optimization may be required to maintain efficient system performance. Additionally, although the system digitizes several recruitment activities, final candidate selection and recruitment decisions may still require manual evaluation by recruiters.

5.3 Future Scope

The future development of TECHXHIRE can focus on integrating artificial intelligence, machine learning, advanced analytics, mobile accessibility, automated communication, and cloud technologies to make the platform more intelligent and efficient for both candidates and recruiters. AI-based job recommendation systems using machine learning algorithms can be implemented to recommend suitable job opportunities to candidates based on their skills, education, experience, interests, and previous applications. An AI-based resume screening system can be developed to extract candidate skills and automatically compare resumes with job requirements. Machine learning techniques can be used to calculate compatibility between candidate profiles and job descriptions and provide candidate-job matching scores. An intelligent chatbot can be integrated to assist candidates with job searches, application-related queries, profile management, and recruitment information. A dedicated Android/iOS mobile application can be developed to provide candidates and recruiters with convenient mobile access to the platform. Automated email, SMS, and push notifications can be implemented for application updates, interview schedules, job recommendations, and recruiter communication. Advanced analytics dashboards with visualization can be introduced to provide recruiters with insights into applications, candidate trends, job performance, and recruitment statistics. The application can be deployed on cloud platforms to improve scalability, availability, and accessibility. Additional security mechanisms such as multi-factor authentication, enhanced encryption, security monitoring, and improved access controls can be implemented to strengthen the platform's security posture.

5.4 Summary

Chapter 5 concluded the report with a comprehensive summary of the project's achievements, limitations, and future scope. The conclusion emphasized the successful development of a centralized recruitment platform and the practical experience gained during the internship. The limitations identified areas for improvement and highlighted the current scope boundaries of the system. The future scope outlined potential enhancements to make the platform more intelligent and efficient through the integration of artificial intelligence, machine learning, mobile accessibility, automated communication, advanced analytics, cloud deployment, and enhanced security mechanisms, thereby paving the way for a more sophisticated and comprehensive recruitment solution.

REFERENCES

Spring Boot Documentation – Spring Framework and Spring Boot Reference Guide. Available at: Spring Boot Documentation

• Spring Data JPA Documentation – Spring Data JPA Reference Documentation. Available at: Spring Data JPA Documentation

• Spring Security Documentation – Authentication and Authorization with Spring Security. Available at: Spring Security Documentation

• React Documentation – React Library Documentation and Development Guide. Available at: React Documentation

• MySQL Documentation – MySQL Reference Manual. Available at: MySQL Documentation

• Postman Documentation – API Testing and Development Platform. Available at: Postman Documentation

• Maven Documentation – Apache Maven Build Tool Reference. Available at: Apache Maven Documentation

• Git Documentation – Git Version Control System Documentation. Available at: Git Documentation

• MDN Web Docs – HTML, CSS, JavaScript, and Web Development Documentation. Available at: MDN Web Docs

• GeeksforGeeks – Java, Spring Boot, React, MySQL, and Full-Stack Development Tutorials. Available at: GeeksforGeeks

• Node.js Documentation – Node.js Runtime and Development Documentation. Available at: Node.js Documentation

• GitHub Documentation – Source Code Management and Collaboration Documentation. Available at: GitHub Documentation

• SSMT_SOLUTION – Internship project requirements, technical discussions, development guidance, and project-related information accessed during the internship.

• TECHXHIRE Project – Project-specific requirements, implementation details, database design, API development, testing, and application workflows developed during the internship.

APPENDICES

Appendix A: Project Code Snippets

A.1 User Entity (Java)

User.java

```java
@Entity
@Table(name = "users")
public class User {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    @Column(unique = true, nullable = false)
    private String email;

    @Column(nullable = false)
    private String passwordHash;

    @Column(nullable = false)
    private String fullName;

    @Enumerated(EnumType.STRING)
    private Role role;

    private LocalDateTime createdAt;
    // Getters and Setters
}
```

A.2 Job Entity (Java)

Job.java

```java
@Entity
@Table(name = "jobs")
@Data
@NoArgsConstructor
@AllArgsConstructor
public class Job {

    @Id
    @GeneratedValue(strategy = GenerationType.IDENTITY)
    private Long id;

    private String title;

    private String description;

    private String location;

    private Double salaryMin;

    private Double salaryMax;

    private String jobType;
```

A.3 API Response Structure

Main.java Run

```java
import java.util.List;

public class ApiResponse<T> {

    private boolean success;

    private String message;

    private T data;

    private List<String> errors;

    // Constructors
    public ApiResponse() {
    }

    public ApiResponse(boolean success, String message, T data, List<String>
    errors) {
        this.success = success;
        this.message = message;
        this.data = data;
        this.errors = errors;
```

Appendix B: Testing Documentation

B.1 API Testing with Postman

Test Case	Endpoint	Expected Result	Actual Result
User Registration	POST /api/auth/register	200 OK	PASS
User Login	POST /api/auth/login	200 OK	PASS
Get Jobs	GET /api/jobs	200 OK	PASS
Create Job	POST /api/jobs	201 Created	PASS
Apply Job	POST /api/applications	201 Created	PASS

Appendix C: Screenshots

Figure C.1: Homepage

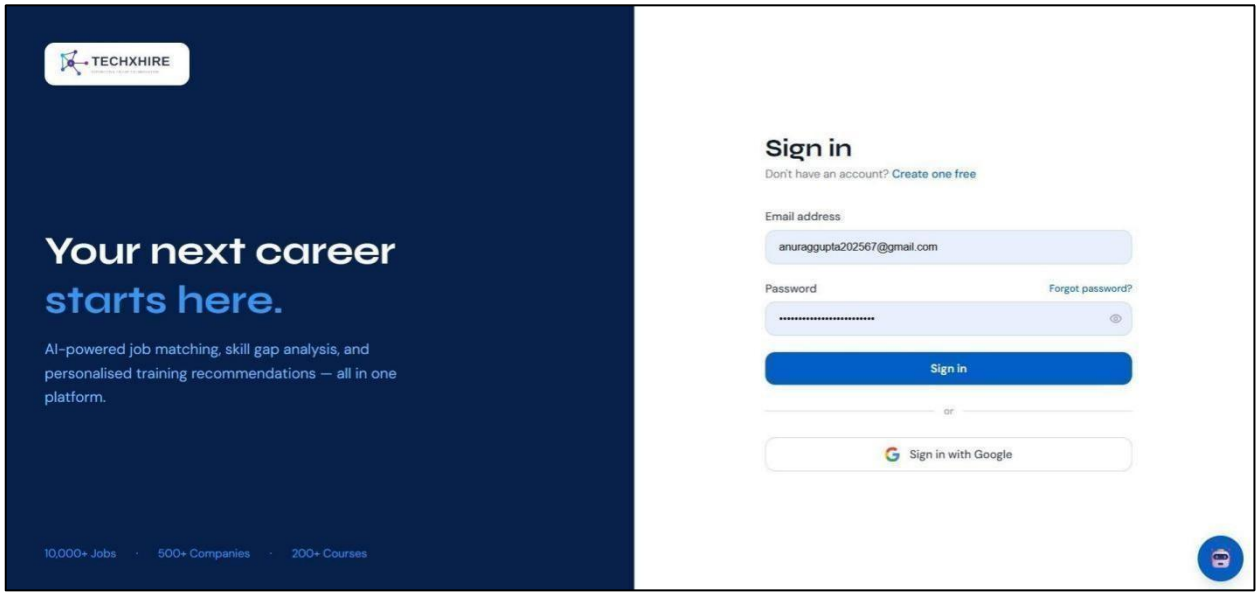


Figure C.2: Job Listing Page

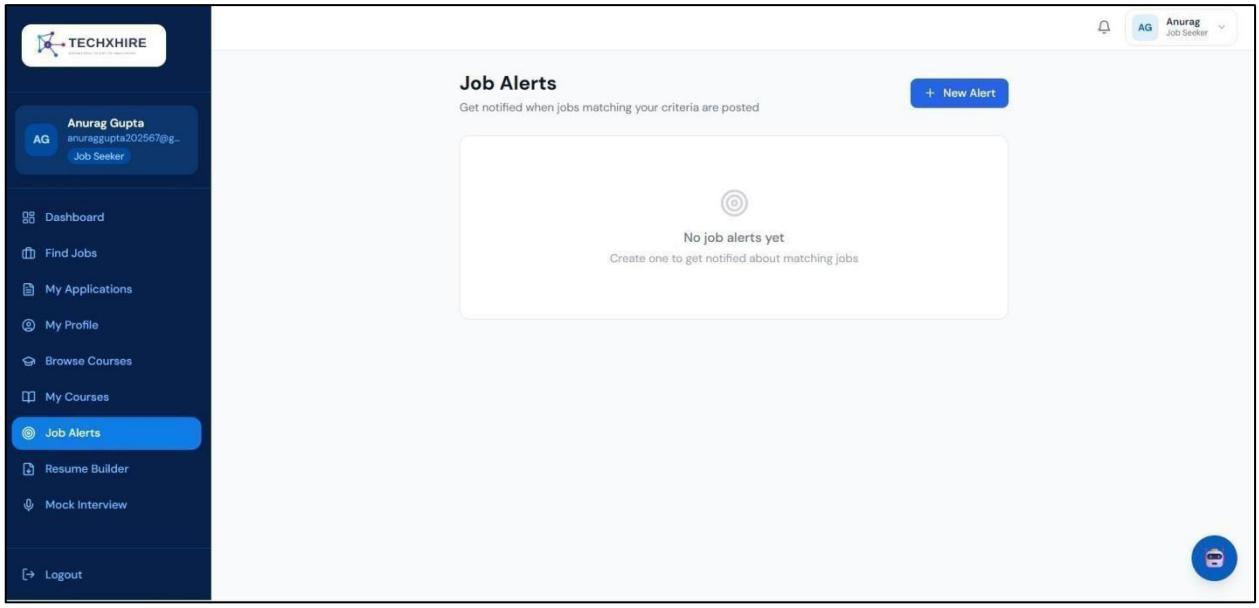


Figure C.3: Candidate Dashboard

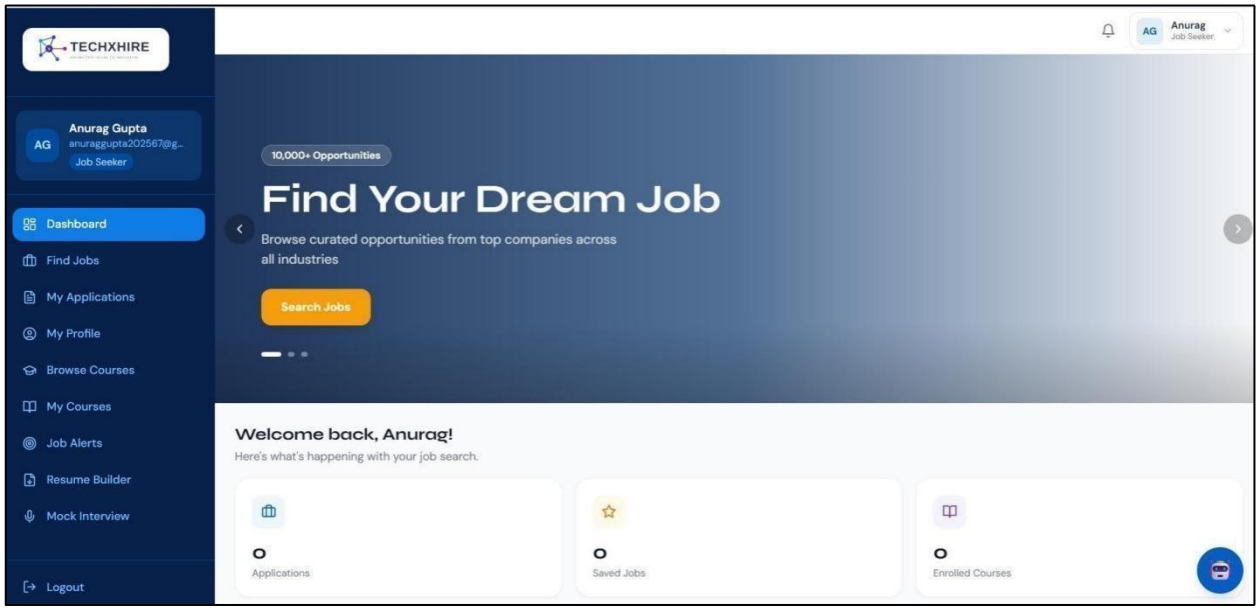


Figure C.4: Recruiter Dashboard

